For Examiner's Use

7 Read the following article and then answer the questions that follow.

Nissan Leaf – the advocate for electric vehicles?

The term electric vehicle (EV) refers to any vehicle that uses electric motors for propulsion. Since 2008, a revival in electric vehicle manufacturing occurred as several national governments established tax rebates, subsidies, and other incentives to promote the introduction of new electric vehicles in the mass market. For example, the current maximum tax credit allowed by the United States Government is US$7,500 per car. Charging an electric car can be done at a variety of charging stations, which can be installed in both houses and public areas.

Currently, there are over 2 million electric cars in use over the world. The Nissan Leaf is the best-selling highway-capable electric car, with over 300,000 units sold globally by January 2018. It is a compact five-door hatchback electric car first introduced in Japan and the United States in December 2010.

The bottom of the car has aerodynamic paneling to reduce drag and improve aerodynamics as much as possible. The Nissan Leaf has a drag coefficient C_d of 0.28 and the official range is 172 km on a full battery charge. The car has a top speed of 150 km h^{-1} and it is suggested that the car is able to go from 0 to 97 km h^{-1} in 9.9 seconds.

The Nissan Leaf uses a 30 kW h lithium-ion battery pack consisting of 48 modules of 4 battery cells each and the *specific energy* of each cell is 140 W h kg^{-1}. The battery pack can be charged using a 230 V 30 A power supply and is expected to retain 70 - 80% of its capacity after 10 years.

The battery capacity percentage P of the Nissan Leaf is given by

$$P = (3.075 \times 10^{-5} x^2 - 0.0064 x + 1) \times 100$$

where x is the number of months of usage. Here, it is assumed that the driver travels 12500 miles on average a year.
(1 mile = 1.609 km)

(a) Suggest an advantage of using electric cars as an alternative to conventional cars using internal combustion engines.

.....

.....

..... [1]

For Examiner's Use

(b) **(i)** Assuming the Nissan Leaf undergoes constant acceleration from rest, determine the distance travelled when it reaches its top speed.

distance = …… m [2]

(ii) The drag force F_d acting on a body is given by

$$F_d = \frac{1}{2} \rho v^2 C_d A$$

where ρ is the density of the medium that the body is moving in, v is the velocity of the body and A is its cross sectional area.

1. Show that C_d has no units.

[1]

2. Given that ρ = 1.2 kg m^{-3} and A = 2.27 m^2, determine the power provided by the engine when the Nissan Leaf is travelling at a constant speed of 50 km h^{-1}.

power = …… W [2]

For Examiner's Use

(c) Fig. 7.1 below shows the energy breakdown for every 100 litres (L) of gasoline consumed by a conventional car that uses an internal combustion engine.

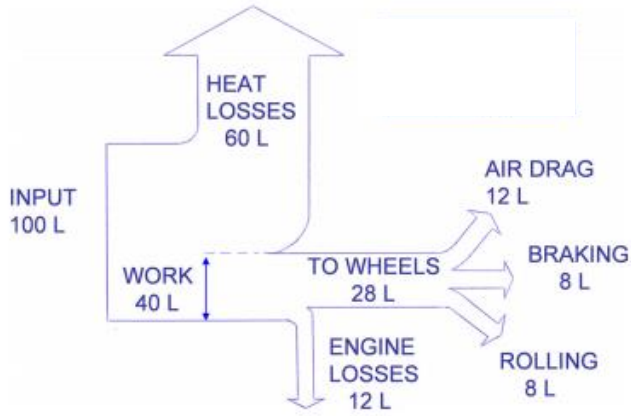

Fig. 7.1

During braking, the kinetic energy of conventional cars is lost to the surroundings as heat generated in the disc brakes due to friction. However, EVs are able to use regenerative braking, where the electric motor acts as a generator, to retrieve part of the kinetic energy lost via braking to charge the car's battery.

(i) State the energy efficiency of the conventional car.

efficiency = …… [1]

(ii) Describe the transformation of energy in the car when regenerative braking is used in EVs.

……

……

…… [1]

(iii) The Nissan Leaf is able to recover 80% of the energy dissipated from braking in the wheels. However, 10% of this energy recovered is lost in the conversion to useful energy stored in the battery pack.

Determine the overall increase in efficiency from **(c)(i)**.

increase in efficiency = …… [2]

For Examiner's Use

(d) **(i)** Deduce what is meant by the term *specific energy*.

……

……

…… [1]

(ii) Determine the mass of each battery cell.

mass = …… kg [1]

(iii) Determine the time taken to fully charge a depleted Nissan Leaf battery pack.

time = …… hours [1]

(iv) Data for battery capacity percentage *P* and number of months of usage x for the Nissan Leaf is given in Fig. 7.2.

x / months	battery capacity percentage *P* / %
0	100.00
12	92.76
24	86.41
36	80.95
48	76.36
60	72.67
72	
84	67.94
96	66.90

Fig. 7.2

1. Complete Fig. 7.2 for x = 72. [1]

For Examiner's Use

The variation with x of P is shown in Fig. 7.3.

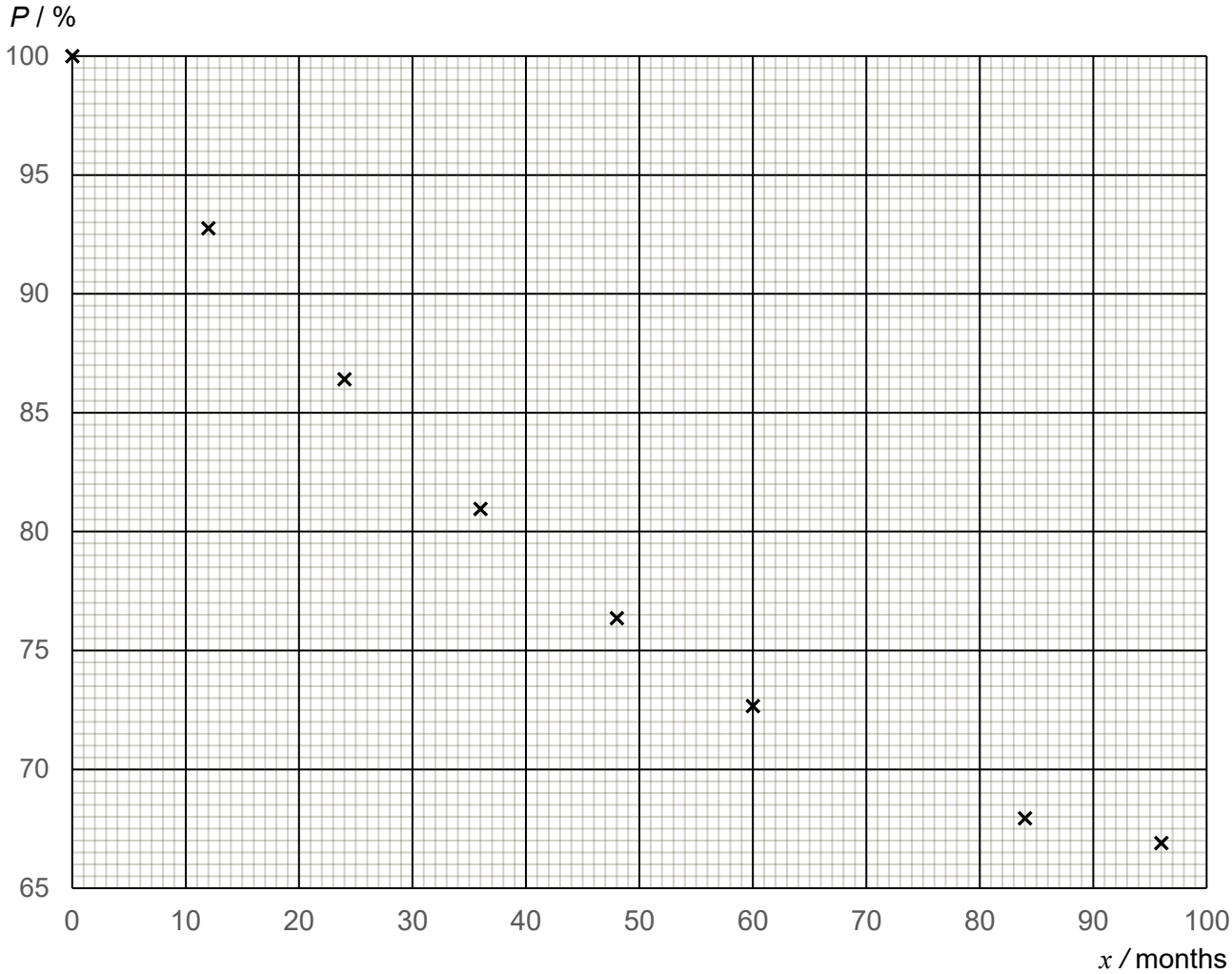

Fig. 7.3

2. Plot the point for x = 72 on Fig. 7.3. [1]

3. Complete Fig. 7.3 by drawing the best-fit line. [1]

4. Use Fig. 7.3 to determine the battery capacity percentage after 45000 miles.

P = …… % [1]

For Examiner's Use

(e) As part of Singapore's efforts to improve air quality, the Land Transport Authority (LTA) announced the introduction of the Vehicular Emissions Scheme (VES). It calculates tax rebates or surcharges for new cars based on the amount of pollutants they produce. The emission factor used for computing the carbon dioxide (CO_2) emissions of EVs is fixed at 0.4 g CO_2 emitted per W h of energy of the EV.

Fig. 7.4 below shows the VES Banding Rebates and Surcharges.

Bands	CO_2 emitted per km travelled (g km^{-1})	Rebate / Surcharge (+/-) for cars ($)
A1	A1 ≤ 90	- 20000
A2	90 < A2 ≤ 125	- 10000
B	125 < B ≤ 160	0
C1	160 < C1 ≤ 185	+ 10000
C2	C2 > 185	+ 20000

Fig. 7.4

(i) Suggest why the emission factor is used for EVs even though they do not produce CO_2.

……

……

…… [1]

(ii) Determine the VES Banding of the Nissan Leaf.